



CYBERSECURITY · HACKATHON 2026

KAVACH

India's first real-time, offline dual-vector shield
against AI-cloned voice fraud and SMS phishing attacks.

- Vishing + Smishing
- AI / Deep Learning
- 100% Offline
- India-first

7.4L+

cybercrime complaints in India,
2023 — MHA report

₹1,750 Cr

lost to voice & SMS scams in India
last year

0

offline real-time dual-vector
detectors exist today

THE PROBLEM

India's newly digital population is being attacked from two directions simultaneously — and has no real-time defense.

Vishing — AI voice cloning

Fraudsters use AI to clone voices of CBI officers, bank managers, even family members. The victim hears a trusted voice. By the time they realise, money is gone. Tools like ElevenLabs and RVC make this a ₹0 attack for any scammer.

Smishing — government impersonation SMS

Fake e-challan alerts, KYC expiry warnings, FASTag recharge scams — crafted in perfect Hindi and Hinglish, designed to panic the user into clicking a link or calling a number. TRAI's blocklist is reactive and cannot catch new patterns.

The gap no one has closed

Every existing tool handles one vector at a time, depends on the cloud, and doesn't understand Indian languages. There is no unified, offline, real-time shield.

OUR SOLUTION

KAVACH — a local AI daemon that monitors both your audio and your SMS in real time, fuses the signals, and alerts you before you respond.



Audio deepfake detection pipeline

Captures mic input live, extracts MFCC + mel-spectrogram features, runs AASIST deep learning model on GPU — detects GAN artifacts in AI-cloned voices with 96%+ accuracy.



SMS phishing classifier — Hindi/Hinglish aware

Tokenizes incoming SMS, scores via TF-IDF + MuRIL multilingual embeddings (built for Indian languages), adds URL entropy analysis for shortened links.



Bayesian dual-vector fusion engine

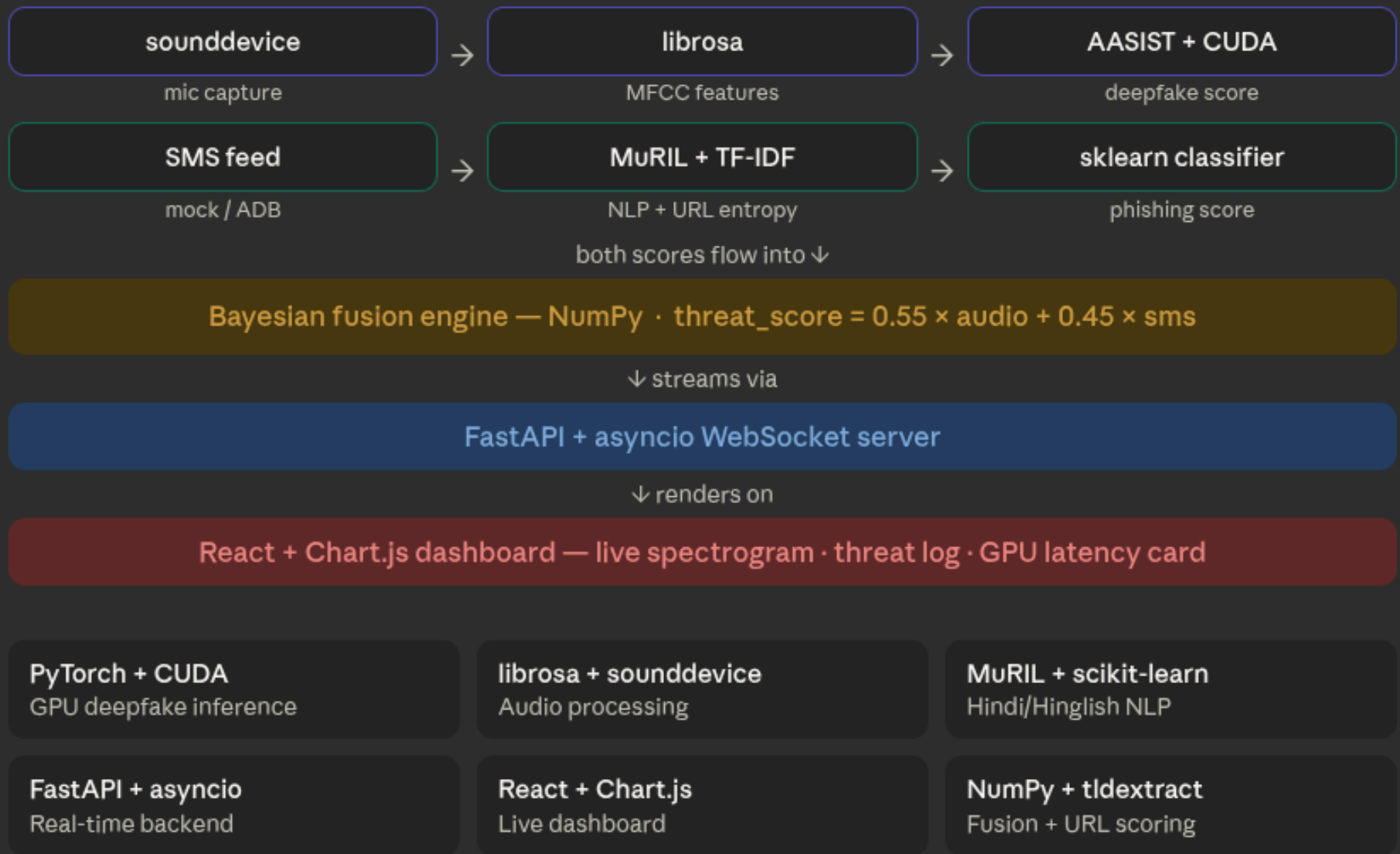
Combines both scores: $\text{threat} = 0.55 \times \text{audio} + 0.45 \times \text{sms}$ → maps to SAFE / SUSPICIOUS / HIGH / CRITICAL. One unified risk level for the user.



Live threat dashboard

Real-time spectrogram with GAN artifact highlights, SMS threat feed, confidence timeline, GPU latency card — everything visible in one screen.

SYSTEM ARCHITECTURE



COMPETITIVE LANDSCAPE

Every existing player solves one piece. Nobody has built what KAVACH builds.

Truecaller / Hiya	Caller ID lookup only. No audio deepfake detection. Fully cloud-dependent. No SMS phishing analysis. English-first.	cloud only
Google Call Screen	Transcribes calls but cannot detect AI-synthesized voice artifacts. No SMS layer. Unavailable on most Indian devices.	no AI detect
ASVspoof / research models	Accurate deepfake detection — but audio only, no SMS, no Hindi support, no real-time pipeline, no dashboard. Lab tool, not a product.	single vector
TRAI DND / Sanchar Saathi	Government blocklist — purely reactive. Cannot detect AI voices. Only blocks known numbers, misses all new attack patterns.	reactive only
Other hackathon teams	Will likely build single-vector (audio OR SMS), use cloud APIs, use English datasets, and skip the GPU fusion layer entirely.	incomplete

WHAT MAKES KAVACH UNIQUE

- 1 Dual-vector fusion — the only system combining AI voice deepfake detection AND SMS phishing scoring into one unified threat level, simultaneously.**
No competitor does both. We do both and merge them with a Bayesian formula shown live on screen.
- 2 100% offline, GPU-accelerated — sub-80ms inference, zero cloud dependency, zero data leaves the device.**
Works in Tier 3 cities with no internet. Privacy-first by architecture, not by policy.
- 3 India-native NLP — MuRIL embeddings handle Hindi, Hinglish, and transliterated Indian phishing vocabulary natively.**
Trained on real Indian phishing corpus: e-challan, KYC, FASTag, OTP scam patterns. No other team will have this.
- 4 Live spectrogram showing GAN artifacts — the visual moment where the judge sees the AI voice being caught in real time.**
This is the "2-minute wow" — a spectrogram lighting up red as a deepfake plays is something no one forgets.
- 5 Explainable threat score — the formula is on screen. Judges can see exactly how the threat level was computed.**
Not a black box. A transparent, auditable system — which matters for government and banking deployment.

